

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)

Enterprise Java

Module - II

Practical – IX

Roll No.:T032	Name:KHUSHI
Class:TYIT	Batch:2
Date of Assignment: 12/09/26	Date/Time of Submission: 12/09/26

9. Implement the following JPA applications.

a. Develop a simple Inventory Application Using JPA.

CheckInventory.jsp

```
<%@page import="java.util.Iterator"%>
<%@page import="java.util.List"%>
<%@page import="javax.persistence.Persistence"%>
<%@page import="javax.persistence.EntityManager"%>
<%@page import="javax.persistence.EntityManagerFactory"%>
<%@page import="javax.persistence.EntityTransaction"%>
<%@page import="myAPP.Inventory"%>
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<%
EntityManagerFactory eMF =
    Persistence.createEntityManagerFactory("INVJPAPU");
EntityManager eM = eMF.createEntityManager();
EntityTransaction eT = null;
List<Inventory> inventoryList;
String submit = request.getParameter("AddToInventory");
if (submit != null && submit.equals("AddToInventory")) {
    try {
        String itemname =
            request.getParameter("itemname");
        String description =
            request.getParameter("description");
        String quantity1 =
            request.getParameter("quantity");
        int quantity =
            Integer.parseInt(quantity1);
        String addedDate =
            new java.util.Date().toString();
        Inventory inv = new Inventory();
        inv.setItemName(itemname);
        inv.setDescription(description);
        inv.setQuantity(quantity);
        inv.setAddedDate(addedDate);
        eT = eM.getTransaction();
        eT.begin();
        eM.persist(inv);
        eT.commit();
    } catch (RuntimeException e) {
        if (eT != null && eT.isActive()) {
            eT.rollback(); }
        throw e; }}
inventoryList =
    eM.createQuery("SELECT i FROM Inventory i",
        Inventory.class)
        .getResultList();
eM.close();%>
<!DOCTYPE html>
```

```

<html>
<head>
  <meta charset="UTF-8">
  <title>Inventory</title>
</head>
<body>
<table border="1">
<tr>
  <td colspan="5" align="center">
    <b>
      Click
      <a href="index.jsp">here</a>
      to add items in the Inventory.
    </b>
  </td>
</tr>
<tr>
  <th>Item ID</th>
  <th>Item Name</th>
  <th>Description</th>
  <th>Quantity</th>
  <th>Added Date & Time</th>
</tr><%
Iterator<Inventory> iterator =
  inventoryList.iterator();
while (iterator.hasNext()) {
  Inventory obj =
    iterator.next();%>
<tr>
  <td><%=obj.getItemID()%></td>
  <td><%=obj.getItemName()%></td>
  <td><%=obj.getDescription()%></td>
  <td><%=obj.getQuantity()%></td>
  <td><%=obj.getAddedDate()%></td>
</tr><%}%>
</table>
</body>
</html>

```

Index.jsp

```

<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head>
  <meta charset="UTF-8">
  <title>Inventory</title>
</head>
<body>
<form action="checkInventory.jsp" method="post">
<table>
<tr>
  <td colspan="2" align="center">
    <b>Add the item in Inventory</b>
  </td>
</tr>
<tr>
  <td>Item Name:</td>
  <td>
    <input name="itemname" maxlength="25" size="50">
  </td>
</tr>
<tr>
  <td>Description:</td>

```

```

<td>
    <textarea rows="5" cols="36"
        name="description"></textarea>
</td>
</tr>
<tr>
    <td>Quantity:</td>
    <td>
        <input name="quantity" maxlength="4" size="5">
    </td>
</tr>
<tr>
    <td colspan="2">
        <input type="submit"
            name="AddToInventory"
            value="AddToInventory">
    </td>
</tr>
</table>
</form>
</body>
</html>

```

Inventory.jsp

```

package myAPP;
import javax.persistence.Column;
import javax.persistence.Entity;
import javax.persistence.GeneratedValue;
import javax.persistence.GenerationType;
import javax.persistence.Id;
import javax.persistence.Table;
@Entity
@Table(name = "inventory")
public class Inventory { @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    @Column(name = "itemid", unique = true, updatable = false)
    private Integer itemid;
    @Column(name = "itemName")
    private String itemName;
    @Column(name = "description")
    private String description;
    @Column(name = "quantity")
    private Integer quantity;
    @Column(name = "addedDate")
    private String addedDate;
    public Inventory() { }
    public Integer getItemID() {
        return itemid;
    }
    public void setItemID(Integer itemid) {
        this.itemid = itemid;
    }
    public String getItemName() {
        return itemName;
    }
    public void setItemName(String itemName) {
        this.itemName = itemName;
    }
    public String getDescription() {
        return description;
    }
    public void setDescription(String description) {
        this.description = description;
    }
    public Integer getQuantity() {
        return quantity;
    }
    public void setQuantity(Integer quantity) {
        this.quantity = quantity;
    }
    public String getAddedDate() {

```

```

return addedDate;}
public void setAddedDate(String addedDate) {
    this.addedDate = addedDate;}}

```

OUTPUT:

The screenshots illustrate the steps to configure a database connection in NetBeans:

- New Persistence Unit:** Shows the 'Provider and Database' tab with 'Persistence Unit Name' set to 'INVJPAAPU2' and 'Persistence Provider' set to 'EclipseLink (JPA 2.1)(default)'.
- Create Data Source:** Shows the 'JNDI Name' as 'jdbc/inventory' and the 'Database Connection' as 'jdbc:derby://localhost:1527/sample [app on APP]'.
- New Connection Wizard (Locate Driver):** Shows the selection of 'MySQL (Connector/J driver)' from the driver list.
- New Connection Wizard (Customize Connection):** Shows the configuration for the MySQL connection:
 - Driver Name: MySQL (Connector/J driver)
 - Host: localhost
 - Port: 3306
 - Database: Inventory
 - User Name: root
 - Password: (masked)
 - Remember password: ☐
 - Test Connection: (button)
 - JDBC URL: jdbc:mysql://localhost:3306/Inventory?zeroDateTimeBehavior=CONVERT_TO_NULL
 - Connection Succeeded: (message)
- Web Browser:** Shows the application running at 'localhost:8080/INVJPA/'. The page title is 'Add the item in Inventory'. It contains a form with fields for 'Item Name' (laptop), 'Description' (Dell laptop), and 'Quantity' (10). A button 'AddToInventory' is present.
- Web Browser:** Shows the application running at 'localhost:8080/INVJPA/checkInventory.jsp'. It displays a table with the following data:

Item ID	Item Name	Description	Quantity	Added Date & Time
1	laptop	Dell laptop	10	Fri Sep 11 13:46:22 IST 2026
2	laptop	Dell laptop	10	Fri Sep 11 13:47:06 IST 2026